

A Novel Sensor Placement Strategy for an IoT-Based Power Grid Monitoring System

Joe-Air Jiang[✉], Senior Member, IEEE, Jen-Cheng Wang[✉], Hung-Shuo Wu,
Chien-Hsing Lee[✉], Senior Member, IEEE, Cheng-Ying Chou[✉], Member, IEEE,
Li-Cheng Wu, Senior Member, IEEE, and Yu-Cheng Yang

Abstract—Dynamic thermal rating (DTR) is a technique that can effectively reduce the complexity of the decision-making processes for a smart grid. Internet-of-Things-based DTR monitoring systems can be used to achieve reliable and low-cost remote monitoring of power grids, but this method is heavily reliant on collecting accurate real-time meteorological data by sensors deployed on the power lines. However, deploying sensors on each span of the line may not be feasible due to the high cost of such sensors. Thus, this article proposes a modified binary particle swarm optimization (MBPSO) strategy to solve multiobjective combinational decision problems. The proposed method is able to determine the minimum number of sensors required to achieve nearly ideal performance. A 161-kV line located between Xizhi and Minquan, part of the Taiwan Power Company transmission system, was selected as the experimental target. The MBPSO algorithm was solved based upon hourly meteorological data provided by Taiwan's Central Weather Bureau. The results obtained with the proposed method show that only one sensor needs to be deployed on the third span of the line to effectively monitor more than 89.6% of high conductor temperature events, and the root mean square error on the reconstructed conductor temperature distribution is less than 0.8 °C.

Index Terms—Binary particle swarm optimization (BPSO), dynamic thermal rating (DTR), Internet of Things (IoT), smart grid.

Manuscript received October 23, 2019; revised March 27, 2020; accepted April 23, 2020. Date of publication April 30, 2020; date of current version August 12, 2020. This work was supported in part by the Ministry of Science and Technology; and in part by the Council of Agriculture of the Executive Yuan, Taiwan, under Grant MOST 108-2321-B-002-037, Grant MOST 108-2811-B-002-510, Grant MOST 108-2622-E-002-023-CC2, Grant MOST 108-2221-E-002-090, Grant MOST 109-2321-B-002-043, Grant MOST 109-2811-B-002-521, Grant 108AS-13.2.11-ST-a5, Grant 108AS-16.2.1-FD-Z2, Grant 109AS-11.3.2-ST-a2, Grant 109AS-14.2.1-FD-Z2, and Grant 109AS-11.3.2-ST-a8. (Corresponding author: Joe-Air Jiang.)

Joe-Air Jiang is with the Department of Biomechanics Engineering, National Taiwan University, Taipei 10617, Taiwan, and also with the Department of Medical Research, China Medical University Hospital, China Medical University, Taichung 40447, Taiwan (e-mail: jajiang@ntu.edu.tw).

Jen-Cheng Wang, Hung-Shuo Wu, Cheng-Ying Chou, and Yu-Cheng Yang are with the Department of Biomechanics Engineering, National Taiwan University, Taipei 10617, Taiwan (e-mail: f98631002@ntu.edu.tw; r05631029@ntu.edu.tw; chengying@ntu.edu.tw; d06631003@ntu.edu.tw).

Chien-Hsing Lee is with the Department of Systems and Naval Mechatronic Engineering, National Cheng Kung University, Tainan 701, Taiwan (e-mail: chienlee@mail.ncku.edu.tw).

Li-Cheng Wu is with the Taiwan Power Research Institute, Taiwan Power Company, New Taipei 238, Taiwan (e-mail: u026942@taipower.com.tw).

Digital Object Identifier 10.1109/JIOT.2020.2991610

NOMENCLATURE

Symbol	Description
q_c	Convection heat loss rate per unit length.
q_r	Radiated heat loss rate per unit length.
q_s	Heat gain rate from the sun.
T_c	Conductor temperature.
$R(T_c)$	AC resistance of conductor at temperature T_c .
I	Conductor current.
$k_{\max}$	Maximum number of iterations.
N	Number of total particles.
P	Number of total spans.
M	Number of total time points.
X_{np}^k	p th span of the n th particle at the k th iteration.
w	Inertia weight.
$pbest_n$	n th individual best particle.
$gbest$	Global best particle.
c_1, c_2	Social parameters set by user.
r_1, r_2	Random numbers between 0 and 1.
d_g	Hamming distance ratio from X_n^k and $gbest$.
d_p	Hamming distance ratio from X_n^k and $pbest_n$.
$ispr$	Ideal sensor placement ratio.
R	Percentage of the lowest ampacity occurred at each span during a period.
T_{mp}	Estimated conductor temperature of the p th span under the lowest ampacity at the m th time point.
c_m	Index that determines whether the high-temperature event is effectively monitored at the m th time point.

I. INTRODUCTION

WITH the rapid growth in the demand for electricity, not only is the development of renewable energy booming but traditional power systems must also be expanded or improved through innovation to face today's highly competitive energy market. How to maintain the reliability and stability of power services may become an even more important issue in the future. When an electric utility provider is not able to supply enough electricity to meet the demand, electricity rationing may be necessary. In Taiwan, electricity rationing due to inadequate infrastructure has been a common occurrence since 1988 [1]. Although the total electricity generated by the Taiwan Power Company (TPC) has gradually increased through the expansion of the grid over the past 30 years, the power supply has already once fallen below the

warning criteria in the past five years, resulting in a major blackout that affected the whole island on August 15, 2017 [2]. Increasing the amount of transmission capacity can be an alternative strategy to reduce risk and vulnerability, which can be achieved by constructing new power grids or realizing the maximum capacity of the existing lines. The former prospect is expensive and time consuming while the latter has been considered the most effective and promising solution in recent years. Dynamic thermal rating (DTR) is one feasible way to maximize the utilization of existing transmission lines [3]. It uses real-time weather information to estimate the maximum allowable current limit of the existing cable, i.e., the ampacity of the transmission line. In general, the ampacity is usually estimated by examining the static thermal rating (STR), under several conservative assumptions [4] but this means that the ampacity is often underestimated. Conversely, DTR provides a way to increase the utilization of transmission lines when weather conditions are favorable, allowing higher conductor loadings [5]. It has been shown that the DTR method can bring huge benefits for the utilization of both overhead transmission lines and underground cables [6].

Currently, the transformation of many power utilities in the world to intelligent cyber-enabled electric power and energy systems (EPES) is underway due to the advancement of the Internet of Things (IoT) [7]. For example, an IoT-based sag monitoring system can be combined with microweather stations to collect real-time weather information to measure and estimate the sag of real transmission lines [8]. In addition to the ambient temperature and wind speed, the conductor temperature and line current also need to be measured in order to determine the actual safety conditions of the transmission line. In fact, the conductor temperature rises rapidly during the peak power supply period. If the conductor temperature exceeds the safety threshold, plastic deformation or the so-called “sag” may occur. Any sag severe enough to violate safety regulations could have disastrous consequences. Excessive thermal stress may even accelerate the aging of the line conductors [9]. In the worst case scenario, such plastic deformation will lead to short circuiting of the transmission lines, perhaps triggering irreversible economic losses. For instance, some areas in North America experienced a massive power cut in 2003 due to power overload, line sag, and the subsequent short circuiting of transmission lines. If the conductor temperature can be measured, the sag problem can be detected early and the sag value of a transmission line can even be estimated [10].

An effective strategy for collecting complete and accurate weather information and transmission line data would be to deploy a sensor on each span of a power line. However, this approach would be costly and does not take into account the excessive use of redundant sensors. The problem can be overcome by solving the sensor placement problem (SPP) that generally considers specific constraints to achieve the lowest cost and highest performance goals. A hybrid algorithm, combining proper orthogonal decomposition (POD) with a genetic algorithm (GA), called the gappy POD-based GA, has been recently proposed to find solutions to the SPP [11]. Transmission efficiency can be improved by

deploying fewer sensors. The estimated ampacity is important information for power dispatch.

The main purpose of this article is to find the optimal sensor placement for a power grid. The ampacity of the extra high-voltage (EHV) transmission lines is first estimated through IEEE Std. 738, which deals with the current–temperature relationship of bare overhead conductors [12]. Then, a monitoring strategy based on the ampacity estimation is proposed. We present a modified binary particle swarm optimization (MBPSO) algorithm to determine the optimal number and location of sensors, and then estimate the ratio of conductor temperature alarms that can be covered by the proposed sensor placement.

II. DYNAMIC THERMAL RATING MONITORING STRATEGY

The STR based upon conductor characteristics and historical meteorological data is usually adopted by power companies to estimate the transmission capacity of the power grid. However, this approach to calculating the ampacity is too conservative so it leads to an underestimation of the true value, which means that the entire grid cannot reach its maximum capacity [13]. An improved strategy proposed in [14], called the seasonal STR strategy, used a set of predetermined static ratings, adjusted for the season or month of the year. However, if the current weather data indicate conditions that are worse than those assumed in determining the STR, it could cause overheating of the lines to overheat and potential damage to the conductors [15]. The DTR based on real-time meteorological data would appear to be a more cost-effective way to help increase the maximum allowable transmission capacity. The DTR method commonly adopts parameters, such as the ambient temperature, conductor temperature, wind direction, wind speed, and solar irradiation. Among these parameters, the most critical factor influencing transmission capacity is the wind speed [16]. Hall and Deb [17] showed the DTR estimated maximum transmission capacity in a mid-latitude climate environment to be up to twice as much as that estimated by STR. However, it should be noted that under low wind speeds and high ambient temperatures the DTR method may overestimate a temperature rise in the transmission lines.

Although the DTR method can increase transmission capacity, the cost associated with monitoring each span of the transmission lines is relatively high, making it impractical to apply in many areas [18]. For a summary of the monitoring strategies and sources of meteorological data in some past studies about DTR (see Table I). For example, Black *et al.* [19] successfully used data from monitoring stations in Northern Ireland and a monitoring strategy makes measurements between two poles and eight locations with lower wind speeds. In contrast, Hosek *et al.* [20] used data for a monitoring station installed on a hillside with high wind speeds. Neither of these studies considered the entire transmission line but only a few spans, which may not include the critical spans or sensor waste. Early DTR studies did not mention how to select monitoring stations [3], [5], [17] while some used official meteorological data instead of *in-situ* measurements [6], [9], [11], [16], [21]. These studies laid the foundation for the SPP, including the incorporation of

TABLE I
SUMMARY OF MONITORING STRATEGIES PROPOSED IN
PAST DTR STUDIES

Authors & reference	Locations	Monitoring stations	Monitoring strategies
Black <i>et al.</i> [19]	Ireland	10	Lower wind speed ^a
Hosek <i>et al.</i> [20]	Canada	1	Higher wind speed ^b
Shaker <i>et al.</i> [3]	Iran	1	Not mentioned
Foss <i>et al.</i> [5]	U.S.A.	3	Not mentioned
Hall and Deb [17]	U.S.A.	1	Not mentioned
Safdarian <i>et al.</i> [6]	Finland	0	External data ^c
Musilek <i>et al.</i> [9]	Canada	0	External data ^c
Kim <i>et al.</i> [16]	Korea	0	External data ^c
Pytlak and Musilek [21]	Canada	0	External data ^c
Matus <i>et al.</i> [18]	Chile	0	Numerical simulation ^d
Jiang <i>et al.</i> [11]	Taiwan	0	Numerical simulation ^d

^aMeasured at two poles and eight locations with lower wind speeds.

^bMeasured on a hillside with strong winds.

^cData from official meteorological agencies.

^dSimulation of algorithms for determining optimal monitoring strategy.

meteorological data and line ampacity simulation. However, due to the high cost of high voltage environmental measurement equipment, there is inevitably a lack of meteorological data from actual measurements. For example, the studies of [6], [9], [16], and [21] used external data of official meteorological agencies to estimate the DTR of the transmission lines. However, the locations of the electric tower are not usually on-site at the measurement station of the meteorological climate grid. In the past, the weather data from neighboring climate grid were usually used to determine the line temperature of the targeted span, which resulted in the estimation error of the line temperature. Besides, these studies were based only on simulations and no real experimentations were carried out. On the other hand, Matus *et al.* [18] developed a heuristic algorithm based on the cable thermal capacity calculation to identify critical spans. In the most primitive form of a heuristic algorithm developed by Matus *et al.*, the placement of the DTR sensor was located at critical spans, which were selected simply by spacing the sensors evenly along the transmission line. Such a placement method is known as the equidistant placement (EDP) strategy. Matus *et al.* also used the probability distribution method to offer a fairly accurate approximation and the amount of data required to build this type of model is greatly reduced. However, decisions for determining the critical spans are based on expert knowledge and unwritten “rule-of-thumb” rules [18]. Obviously, this method ignores the variability of weather conditions and the changes of terrain along the entire transmission line so that it also still requires a large number of sensors in order to be useful. Especially, there can be considerable variation in weather conditions on different parts of the lines [18]. This approach, although useful, can quickly escalate into an unmanageable task as the number of spans increases indefinitely. Jiang *et al.* [11] adopted a GA to determine the optimal sensor placement and reconstruct the conductor temperature distribution. They presented the simulation results, but also lacked for measuring the actual temperature of the transmission line to verify the coverage performance of sensor placement [11]. Different from the above studies, in this article, we calculate the ampacity and conductor temperature of 161-kV transmission lines

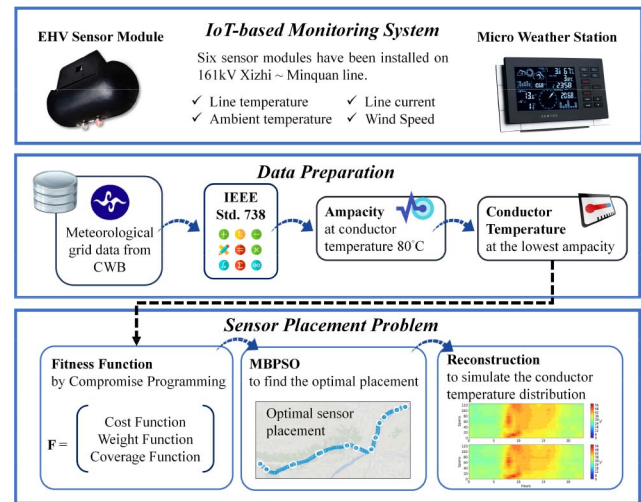


Fig. 1. IoT-based monitoring architecture for transmission lines as proposed in this article.

to determine the best sensor placement to effectively monitor high-temperature events. In addition, we also actually measure the line temperature by using the EHV sensor node, as a means of verifying the efficacy of the proposed method. However, the results show that the wind speed at the electric tower was not measured correctly; the data will be discussed in greater detail in Section IV.

Fig. 1 shows the IoT-based monitoring architecture for transmission lines proposed in this article. A numerical simulation is used to find the optimal sensor placement for the installation of the designed EHV sensor modules on the 161-kV transmission line between Xizhi and Minquan. In the data preparation stage, the IEEE Std. 738 was used to calculate the ampacity at 80 °C, based on meteorological data from the CWB in Taiwan. Next, it is necessary to find the lowest ampacity of all spans for calculation of the conductor temperature distribution. According to the proposed MBPSO algorithm, the simulated conductor temperature is one of the optimized adaptation functions needed to find the best sensor deployment. The POD method elaborated in [11] was then used to verify whether the reconstructed conductor temperature distribution is reasonable.

The conductor temperature and line ampacity are dependent upon weather conditions. Several standards, such as IEC, CIGRE, and IEEE, have been suggested for modeling of the conductor temperature in relation to weather conditions [12], [22], [23]. These models are based on the assumption of the steady-state thermal equilibrium and that the heat gain is equal to the heat loss. In addition, the meteorological parameters affecting the thermal state of the conductor, including the average wind speed, wind direction, turbulence, ambient temperature, and solar radiation, are taken into account. Compared with CIGRE, the IEEE standard has lower computational complexity, but the difference between the two for the estimation of the DTR of the transmission lines is usually not significant [24], [25]; on the other hand, the IEC model has been found to be simpler to use but less accurate than the IEEE or CIGRE models [26]. After considering

the required accuracy and the computational burden, we chose the IEEE standard for calculating the conductor temperature and ampacity. According to the IEEE Std. 738-2012, the conductor temperature can be estimated based on a heat balance principle of heat gains and losses in the conductor, i.e.,

$$q_s + I^2 \cdot R(T_c) = q_c + q_r. \quad (1)$$

Equation (1) represents the steady heat balance equation; other details are defined in the IEEE Std. 738-2012. In general, the weather conditions and current flows of overhead transmission lines are time variant. Although the transient rating calculation is subject to varying current conditions, weather conditions are usually assumed to be constant. In fact, thermal modeling using the average wind speed and air temperature is a much better choice than using instantaneous values [20]. The conductor temperature can be calculated under steady-state conditions by using a specified current and a time-varying meteorological factor for a specific time.

III. MODIFIED BINARY PARTICLE SWARM OPTIMIZATION

The SPP for determining optimal sensor placement for an IoT-based monitoring system usually focuses on identifying the minimum number of sensors needed to obtain the highest accuracy. This involves solving an integer selection problem with nonpolynomial (NP) complexity. The ideal choice of sensor placement locations and numbers [27] may be unsolvable. Heuristic algorithms have been commonly used in various optimization methods in recent years. In most of the past sensor selection schemes mentioned in the literature, heuristic algorithms have been utilized to obtain optimal solutions [28]. The cost of the sensors and the accuracy of the measurement must be simultaneously considered in the SPP since the cost of installing these specialized sensors in a transmission system for a power grid is very expensive, especially in an EHV environment. Although heuristic algorithms have been widely used in sensor placement optimization, their performance in solving SPPs under given different cases is inconclusive. The most commonly used among the various heuristic algorithms is the GA. For example, in the research done by Worden and Burrows [29], a GA, simulated annealing (SA), and exhaustion method were used to examine the optimal locations of sensors for fault detection. GAs are the methods with computational intelligence to effectively achieve global optimization during solving the SPP [30]. GAs have also been applied to structural health monitoring systems and spatial road network SPPs [30], [31]. In spite of many merits of GAs, GAs still suffer from various difficulties in designing the objective function, getting the right representation of operators, reducing computational burden, etc.

Particle swarm optimization (PSO) is also a heuristic algorithm method that was first proposed by Kennedy and Eberhart [32]. It was used to simulate the social behavior of a flock of birds. PSO was originally developed for a continuous-valued problem space, but since then many problems have been defined for a discrete-valued space where the variable domain is finite. Typical examples of these problems are integer programming, scheduling, and routing

problems [33]. Therefore, Kennedy and Eberhart [34] introduced a discrete binary PSO (BPSO) version for solving discrete optimization problems. Although it has been demonstrated in some studies that GA is feasible for SPP, other studies indicate that BPSO may yield better results [28], [35]. Accordingly, an MBPSO algorithm is proposed in this article to obtain a feasible optimal solution within a reasonable calculation time. Here, this article uses an MBPSO algorithm to solve the SPP for EHV transmission lines, attempting to achieve better results and faster convergence speeds.

A. Binary Particle Swarm Optimization

In the study of [32], the PSO considered a group consisting of N candidate solutions, which they called particles. These candidate solutions explored the P -dimensional hyper-space during k iterations via updating the position and velocity of these particles to search for the global optimal solution. During the optimization process, the PSO technique simulated the predation of bird populations within a limited time period as a recursive process. The iterative process was repeated until an acceptable fitness or the maximum number of iterations was reached. In a recent study, Mishra *et al.* [36] used the BPSO technique to reduce the number of substations, where all voltage levels were observable under various practical constraints. Three important formulas from [36] are repeated here for reference as follows:

$$V_n^{k+1} = wV_n^k + c_1r_1(pbest_n - X_n^k) + c_2r_2(gbest - X_n^k) \quad (2)$$

$$\text{sigmoid}(V_{np}^k) = \frac{1}{1 + e^{-V_{np}^k}} \quad (3)$$

$$X_{np}^k = \begin{cases} 1, & \text{if } \text{rand}(0, 1) < \text{sigmoid}(V_{np}^k) \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Velocity updating and position updating are the two primary functions of the BPSO algorithm operations. After iteratively calculating the particle velocity, a sigmoid function is used to determine whether each bit on the particle is 1 or 0, as shown in (2)–(4), where V_n^k and X_n^k represent the velocity and position of the n th particle at the k th iteration, respectively; and $pbest_n$ and $gbest$ indicate the individual best solution and the global best solution, respectively, $0 < n \leq N$, and $0 < k \leq k_{\max}$. The possible solution X_i^k is a string of binary array which represents the location of sensor deployment, with 1 meaning that a sensor should be installed at a specific location and 0 means the opposite. In general, the inertia weight w and social parameters c_1 and c_2 are preset by the user, where $c_1 > 0$ and $c_2 > 0$. The random numbers r_1 and r_2 are randomly given from a uniform distribution between 0 and 1. For a particle, the p th bit indicates whether the sensor is going to be installed at the p th span of the transmission line.

However, some deficiencies in BPSO performance have been revealed in recent studies [37]. For example, the velocity update function indicates that when V_{np}^k is close to zero, $\text{sigmoid}(0)$ approaches 0.5 over time, which means that the particles' bits will change purely randomly. Moreover, when

the position of particles in the PSO is forced to change from continuous to discrete, it lacks the ability to memorize better solutions. As iterations continue, the proportion of random assignments increases and the best solution is rarely updated. In addition, Mirjalili and Lewis [38] showed that BPSO is limited to providing local optimal solutions. As a result, in this article, we redefine the binary decision function in the proposed MBPSO algorithm.

B. Modified Binary Decision Function

The velocity function is not considered in the MBPSO algorithm proposed in this article; rather the particle position is directly updated. The distance between a particle and its individual best solution or global best solution is calculated as in (2). Since the distance between binary arrays cannot be directly subtracted, the Hamming distance is used to determine the difference between two particles. Based on the signaling theory, the Hamming distance between two strings of equal length (e.g., two binary arrays $\mathbf{X}$ and $\mathbf{Y}$) is defined by [39]

$$\text{Hamming}(\mathbf{X}, \mathbf{Y}) = \sum (\mathbf{X} \oplus \mathbf{Y}). \quad (5)$$

This means that the Hamming distance between the two binary arrays is the sum of the two arrays after the XOR operation $\oplus$, i.e., the number of positions for different symbols in $\mathbf{X}$ and $\mathbf{Y}$. Moreover, the Hamming distance ratio of X_n^k from $pbest_n$ and $gbest$ can be calculated by

$$d_p = \text{Hamming}(pbest_n, X_n^k) / P \quad (6)$$

$$d_g = \text{Hamming}(gbest, X_n^k) / P, \text{ respectively.} \quad (7)$$

To prevent the proposed MBPSO algorithm from falling into a local optimal solution, a mutation function is used, which is given by

$$\text{mumate}(X_{np}^k) = \begin{cases} 1, & \text{if } \text{rand}(0, 1) < ispr \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

In (8), $ispr$ refers to the ideal sensor placement ratio, $0 < ispr < 1$, that is, on what proportion of a span are sensors expected to be installed. The random function $\text{rand}(0, 1)$ gives a random number between 0 and 1. Based on the concept of GA, the mutation function mutates particles that satisfy specific conditions to increase the chance of escaping the local optimal solution. Although it is expected that with the SPP, better sensor placement can be achieved using fewer sensors, too few sensors may not be able to effectively monitor the whole transmission line. Hence, a certain proportion of particle length (i.e., $ispr$) is reserved for mutation instead of allowing completely random mutation. By doing so, it is expected to increase the speed of convergence.

A flowchart of the proposed MBPSO algorithm is depicted in Fig. 2. First, in prepreparation, the position of the particle is determined and then used to calculate the Hamming distance ratio. Here, X_{np}^{k+1} means the p th bit in the position string of the n th particle for the $k+1$ th iteration, which is going to be determined. In simple terms, if the particle is far from $gbest$, let the particle be close to $gbest$. Similarly, if the particle is far from $pbest_n$, let the particle be close to $pbest_n$. Finally,

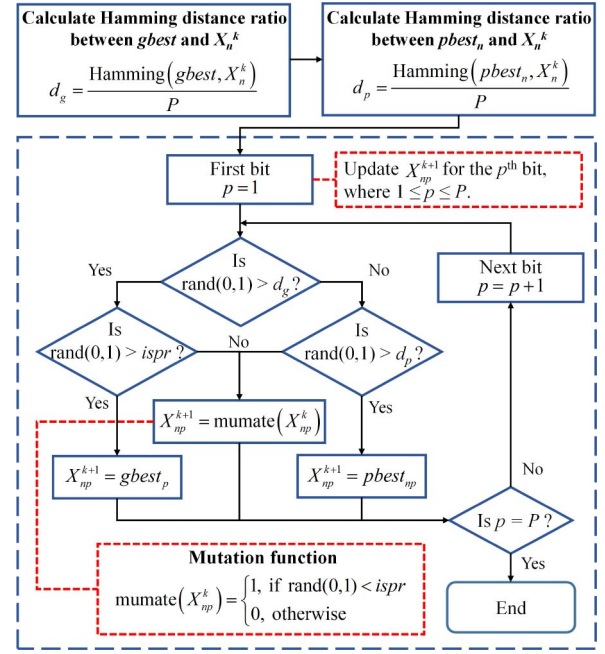


Fig. 2. Flowchart of the proposed MBPSO algorithm.

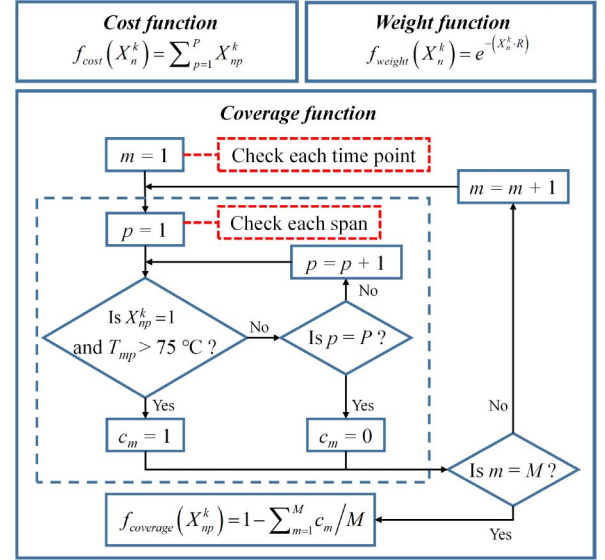


Fig. 3. Formulas and flowchart for the fitness functions.

the mutation function is used to increase the probability of escaping the local executed once for every bit on X_n^{k+1} .

C. Fitness Function

Fitness refers to a criterion that is used to evaluate the excellence of the solution of an optimization algorithm. In order to deal with the multiobjective SPP, this article divides the fitness function into three functions: 1) the cost; 2) the weight; and 3) the coverage functions. Fig. 3 shows the flowchart and formula for the entire fitness function. The cost function f_{cost} is the cost of the sensors. Assuming that the installation costs for each sensor are the same, and f_{cost} can be simplified to the

number of sensors as follows:

$$f_{\text{cost}}(X_n^k) = \sum_{p=1}^P X_{np}^k. \quad (9)$$

Moreover, the weight function f_{weight} refers to the weight given to the span that actually produces the lowest ampacity. During a period of time M , the ampacity for each span of the transmission line is calculated to find the lowest one at each time point. Then, the ratio R of the occurrence of the lowest ampacity for each span during this period is accumulated as a weighted reference. To allow the weight to fall between 0 and 1, the same range used by the cost function, the Euler number e is used to process R , shown as follows:

$$f_{\text{weight}}(X_n^k) = e^{-(X_n^k \cdot R)} \quad (10)$$

where $X_n^k \cdot R$ denotes the inner product of X_n^k and R . If the selected span often produces the lowest ampacity, this inner product is larger.

According to the data sheet for the aluminum conductor steel reinforced (ACSR) 795 MCM cable used in the EHV transmission line, the upper limit of the normal operating temperature for the cable is 80 °C. For a candidate solution X , the coverage function f_{coverage} verifies if the X_n^k can effectively monitor the high-temperature events over a period of time. Here, m represents the time points and $m = 1, 2, \dots, M$, respectively. The ampacity of the transmission line at a critical conductor temperature of 80 °C is first calculated and the lowest ampacity of all spans as a reference for the current boosting is found. Next, the steady-state conductor temperature is calculated by assuming that the current is boosted to the minimum ampacity. If the conductor temperature of any selected span is higher than a warning temperature of 75 °C, it is considered that the sensor placement is suitable to cover high-temperature events at this time point. This means that the conductor temperature for a certain span has approached the critical temperature of 80 °C. Furthermore, this article uses the coverage rate of the above high temperature events for a given sensor placement at a time interval M as a coverage function f_{coverage}

$$f_{\text{coverage}}(X_{np}^k) = 1 - \frac{\sum_{m=1}^M c_m}{M}. \quad (11)$$

As can be seen in Fig. 3, the coverage function needs to check whether the conductor temperature for any selected span in X_n^k is greater than 75 °C at the m th time point. If so, c_m is 1; otherwise, it is 0. However, it is necessary to subtract the coverage rate from 1 to allow f_{coverage} to fall between 0 and 1, the same range used by the cost function.

IV. CASE STUDY AND MEASUREMENT RESULTS

In cooperation with the TPC, sensors were deployed on 161-kV transmission lines. The target transmission line is located between Xizhi and Minquan, which is one of the most heavily loaded lines in northern Taiwan. The selected line has a total of six transmission towers, corresponding to six spans. The exact location of this line cannot be disclosed,

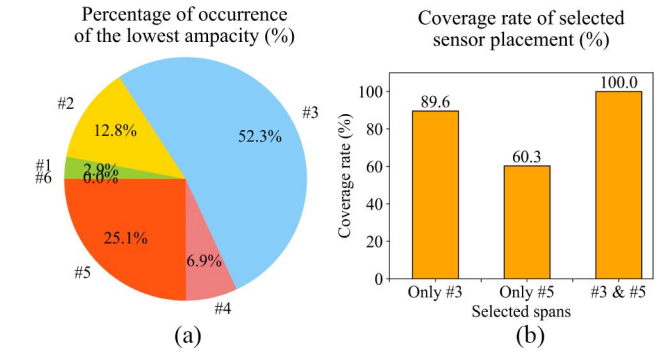


Fig. 4. (a) Percentage of occurrence of the lowest ampacity of all spans. (b) Coverage rate for the sensor placement solution on the Xizhi–Minquan line in 2017.

but it is situated on a hill near Xizhi. Using the meteorological data provided by the CWB and the current data obtained from the TPC in 2017, the estimated ampacity of this line was solved using the IEEE Std. 738 [12].

A. Optimal Sensor Placement for the Xizhi–Minquan Line

Fig. 4(a) shows the accumulated percentage of occurrences of the lowest ampacity on the six spans in 2017. This percentage is equal to the ratio R used for the calculation of the weight function. Results indicate that 52.3% of the lowest ampacity occurs on the third span while the fifth span accounts for 25.1%. The span with the lowest ampacity occurrence on a transmission line is considered to be a thermal bottleneck. Consideration of the location of these thermal bottlenecks should be given for higher priority of sensor deployment. Since the line has only six spans, the climatic conditions are quite similar all along the line. Thus, when the meteorological parameters are similar, the factors that actually affect the conductor temperature or ampacity are the longitude, latitude, altitude, and other geographic parameters along the span. These factors, which originally were assumed to have a smaller effect, produced differences in the location of thermal bottlenecks.

The proposed MBPSO algorithm is used to solve the SPP. The selected spans and the recommended sequence are shown in Fig. 4(b). As expected, the final result is that the third and fifth spans are selected. If only the third span is selected, the coverage rate would be as high as 89.6%. If both the third and fifth spans are chosen, the result shows that the coverage rate reaches 100%. Although this is not a realistic option, it reveals that this sensor placement would offer much better monitoring performance. Accordingly, monitoring of the third span of the Xizhi–Minquan line should be given priority. A photograph of the six sensor modules installed in a location on the third span of this line is depicted in Fig. 5.

B. Measurement Results Obtained by Sensor Modules

In Taiwan, the TPC has adopted a two-conductor three-phase design for the 161-kV circuits. This means that there are a total of six bundles of cables for one transmission line. This corresponds to six EHV sensor modules, which are labeled A1, A2, B1, B2, C1, and C2. The six EHV sensor modules are used

TABLE II
MEASUREMENT RESULTS OF THE XIZHI–MINQUAN LINE AT FIRST EIGHT DAYS FROM DECEMBER 3, 2017

Sensor units	Line temperature (°C)			Ambient temperature (°C)			Line current (A)			Wind speed (m/s)		
	Min.	Ave.	Max.	Min.	Ave.	Max.	Min.	Ave.	Max.	Min.	Ave.	Max.
A1	12.32	16.43	26.82	12.93	16.48	23.01	206.53	256.03	318.60	-	-	-
A2	12.05	15.91	25.41	12.88	16.45	22.71	221.86	274.20	356.22	-	-	-
B1	13.17	17.43	30.46	13.98	17.50	26.69	202.09	251.68	313.95	-	-	-
B2	12.30	16.79	32.34	13.45	17.06	26.03	222.51	277.66	347.44	-	-	-
C1	12.21	15.99	31.42	12.39	16.03	24.67	212.26	261.02	320.97	-	-	-
C2	12.22	16.32	26.70	13.28	16.84	23.66	196.81	244.53	302.69	-	-	-
TPC	-	-	-	-	-	-	226.75	279.84	339.10	-	-	-
CWB	-	-	-	13.40	16.27	20.90	-	-	-	0.40	2.53	5.30
Gateway	-	-	-	-	-	-	-	-	-	0.00	5.64	25.07

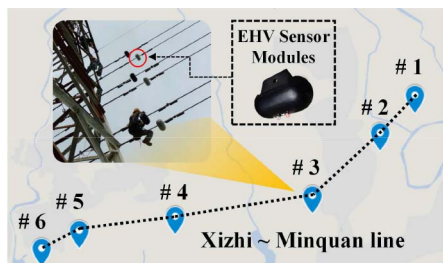


Fig. 5. Six sensor modules installed on the third span of the Xizhi–Minquan line.

to measure the line temperature, ambient temperature, and line current, while the wind speed is measured by a microweather station within the gateway installed at the tower base. The modules were initially installed at 14:00, December 3, 2017 and the sampling rate was set to every 5 s. However, the measurement results were averaged per hour to be compatible with the hourly meteorological data obtained from the CWB.

The measurement results for the first eight days starting December 3, 2017 are plotted in Fig. 6 and tabulated in Table II. For comparison of the accuracy of the measurement, the line current provided by the TPC and the ambient temperature and wind speed provided by the CWB are used as references. It can be observed from the figure that measurements recorded by the six sensor modules are fairly consistent regardless of line temperature, ambient temperature, and line current. According to the mathematical simulation results for the IEEE Std. 738 model, when the wind speed is greater than 1 m/s, the line temperature is more closely related to the ambient temperature and less closely related to the current. Conversely, if the wind speed is less than 1 m/s, the line temperature will rise sharply as the wind speed decreases, reaching the highest temperature at zero wind speed. The measured ambient temperature also matches that provided by the CWB. The line temperature falls between 12 °C and 33 °C and the ambient temperature falls between 12 °C and 27 °C. Obviously, the line temperature is closely related to the ambient temperature, but the peak line temperature is higher than the ambient temperature at the peak temperature on December 10, probably as a consequence of the low wind speed. On December 10, the wind speed is generally lower than 1 m/s in the morning and reaching the highest wind speed of 1.8 m/s at night. Therefore, the line temperature

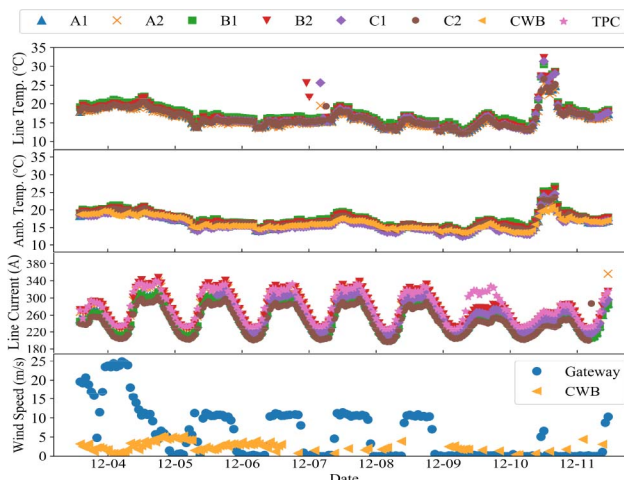


Fig. 6. Line temperature, ambient temperature, line current, and wind speed measured at first eight days from December 3, 2017.

is relatively high with a positive correlation with the ambient temperature. Although the line current is lower than on the previous six days, this is not the main factor that affects the line temperature. The cold and windy winter means that the electricity consumption is relatively low compared to the summer, with the average values of both the line temperature and the ambient temperature being about 16 °C–17 °C. Some line temperature inconsistencies appear on December 7 and 10, indicating potential inaccuracies in the sensor modules or differences between the three phases of the cables. It must be noted that the vertical height difference between each phase is about 6 m, so the resulting difference in wind speed may be the cause of the inconsistency.

The line current falls between 196 and 339 A, with a significant change between day and night; peak electricity usage is between 8:00 and 16:00 each day. The observations show that the current on December 9 was significantly higher than for the sensor modules. The six sensor modules measured a line current of less than 300 A on December 9, but the maximum current was greater than 330 A. Since the six sensor modules are unlikely to have consistent measurement errors, the current measured by the TPC at the two poles might be different from the current measured by the sensor modules on the third span. The exact reason for this difference needs to be investigated further, but the issue has been brought to light.

In addition, the wind speed was measured by the gateway but since this was set at the tower base, the wind speed did

TABLE III
RMSE OF THE RECONSTRUCTED LINE TEMPERATURE

Date	RMSE (°C)				
	03:00	10:00	15:00	21:00	All day
12/03	-	-	0.1825	0.0827	0.1816
12/04	0.2226	0.0655	0.1333	0.0199	0.1677
12/05	0.0625	0.2783	0.1053	0.0486	0.1154
12/06	0.0889	0.2037	0.1408	0.3525	0.1874
12/07	0.1108	0.6235	0.1369	0.0058	0.2919
12/08	0.1804	0.0804	-	-	0.1131
12/09	0.0954	0.1573	0.0532	0.2035	0.1191
12/10	0.7326	0.2474	0.2795	0.0346	0.3751
12/11	0.0609	0.0426	0.0242	0.1045	0.0653

not reflect the actual situation. The measurement results are extremely exaggerated, with an unreasonable value of 25 m/s per hour on the first day occurs. Thus, the gateway wind speed measurement was discarded and will not be discussed. In this article, the wind speed from the CWB is adopted, with data interpolated by the Kriging method after measurement by each CWB observation station. The average wind speed according to the CWB for the first eight days was 2.53 m/s and the maximum value was 5.30 m/s.

C. Reconstruction of Line Temperature

The optimal sensor placement for the Xizhi–Minquan line was previously obtained based on the proposed MBPSO algorithm and the actual data measured at the third span for the first eight days. This section verifies the reliability of the selected sensor placement through the reconstruction of the conductor temperature distribution. Therefore, the conductor temperature distribution is first estimated based on the ambient temperature and wind speed of CWB. The reconstruction of the conductor temperature field is an alternative approach to the coverage rate to understand the effectiveness of the selected sensor placement. In [11], the conductor temperature of selected transmission lines was reconstructed by POD, the reconstruction error was used as the fitness value, and GA was used to solve the optimal sensor placement. The same reconstruction method was applied in this article.

For example, on December 10, the measured line temperature, calculated line temperature, and reconstructed line temperature from the EHV sensor modules on the third span are depicted in Fig. 7. The line temperature was calculated based on the IEEE Std. 738 with the ambient temperature and wind speed from the CWB. The line temperature was reconstructed using the POD method with only the characteristics of the third span retained. Simulations were carried out for four different time points on December 10, 2017, i.e., 03:00, 10:00, 15:00, and 21:00. The root-mean-square error (RMSE) results between the calculated line temperature and reconstructed line temperature for the six spans obtained at the four considered time points and throughout the day are summarized in Table III. The EHV sensor modules were measured from 14:00 on December 3, 2017, but the data loss occurred during the second half of December 8, so the line temperature could not be simulated. Since there is a little difference in the spatial distribution of the line temperature over only six spans, the observed maximum RMSE does not exceed 0.8 °C.

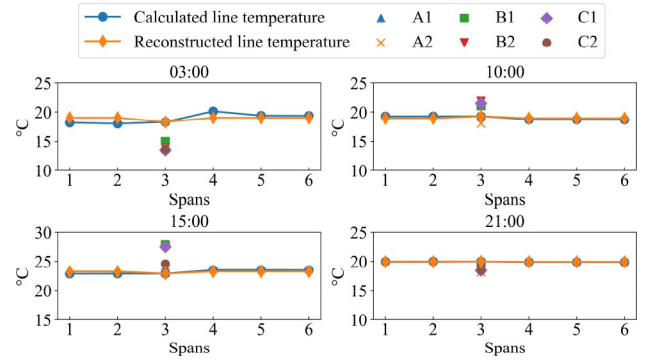


Fig. 7. Measured line temperature, calculated line temperature, and reconstructed line temperature on December 10, 2017.

However, there is an error between the sensed values and the simulated results for the third span. The most likely reason is that the wind speed provided by the CWB is different from the actual wind speed blowing on the transmission line. In fact, the wind speed provided by the CWB is actually an interpolated value, and the height of the measurement is different from the height of the overhead transmission line, so it is only for reference. Still, the CWB measurements can be used as a data source for simulating the optimal sensor placement before installing sensors in practice.

The simulation results are obtained using the CWB grid wind speed, which is actually an average wind speed for a 2.5 km × 2.5 km area. Note that the wind speed provided by the CWB is an interpolated value, and the height of the recorded measurement might be different from the height of the overhead transmission line, so it is only for reference. An inspection of Fig. 7 indicates that there are several biased results among measured, calculated, and reconstructed line temperatures. At 03:00 and 21:00 at night on December 10, 2017, the actual wind speeds on the electric tower might have been stronger than the average wind speeds reported by the CWB, so a lower measured line temperature would be obtained. On the other hand, the actual wind speeds at 10:00 and 15:00 on December 10, 2017 might have been weaker than the reported average CWB wind speeds, resulting in the measurement of a higher line temperature. Compared with the instability of the wind speeds, the measured ambient temperatures are relatively consistent with the average CWB ambient temperatures, which are listed in Table II. The most likely reason is that the wind speeds provided by the CWB are different from the actual wind speeds on the transmission line. In other words, the difficulty in measuring wind speed is the main cause of simulation errors.

V. CONCLUSION

An IoT-based monitoring architecture for the power grid is proposed in this article. An MBPSO algorithm is also proposed to solve the problem of sensor placement on EHV transmission lines based on the estimated DTR. Using the proposed method, we successfully deployed sensors on the third span of the Xizhi–Minquan line. As the TPC system increased the transmission current, the proposed sensor placement is still able to

monitor high-temperature events and give warning before any of the spans reaches the critical operating temperature. Using historical weather data and line current data, it is found that the proposed sensor placement can effectively monitor more than 89.6% of high conductor temperature events when sensors are installed only on the third span of the Xizhi–Minquan line. Moreover, simulating the reconstruction performance of the selected sensor placement through dimensionality reduction and reconstruction method, the RMSE of reconstruction is less than 0.8 °C. The results show that the proposed strategy is able to provide a reliable technology to increase the current capacity of transmission lines and assess the risks of power grids under given operating conditions and measured weather information. In the future, the sensing data will be continuously analyzed to verify whether deploying sensors on the third span matches the expected results.

REFERENCES

- [1] Taiwan Power Company. (2017). *Number of Power Cuts Over Years*. Accessed: Oct. 13, 2018. [Online]. Available: <https://www.taipower.com.tw/tc/page.aspx?mid=214&cid=124&cchk=1314f228-289f-4a7d-b8df-ecb096b672eb>
- [2] *Blackout Report*, Department of Information Services, Executive Yuan. Accessed: Oct. 13, 2018. [Online]. Available: <https://english.ey.gov.tw/Page/61BF20C3E89B856/92c55c68-0871-49fa-8e95-80e1b30d27d4>
- [3] H. Shaker, M. Fotuhi-Firuzabad, and F. Aminifar, "Fuzzy dynamic thermal rating of transmission lines," *IEEE Trans. Power Del.*, vol. 27, no. 4, pp. 1885–1892, Oct. 2012.
- [4] P. M. Callahan and D. A. Douglass, "An experimental evaluation of a thermal line uprating by conductor temperature and weather monitoring," *IEEE Trans. Power Del.*, vol. 3, no. 4, pp. 1960–1967, Oct. 1988.
- [5] S. D. Foss, S. Lin, R. A. Maraio, and H. Schrayshuen, "Effect of variability in weather conditions on conductor temperature and the dynamic rating of transmission lines," *IEEE Trans. Power Del.*, vol. 3, no. 4, pp. 1832–1841, Oct. 1988.
- [6] A. Safdarian, M. Z. Degefa, M. Fotuhi-Firuzabad, and M. Lehtonen, "Benefits of real-time monitoring to distribution systems: Dynamic thermal rating," *IEEE Trans. Smart Grid*, vol. 6, no. 4, pp. 2023–2031, Jul. 2015.
- [7] G. Bedi, G. K. Venayagamoorthy, R. Singh, R. R. Brooks, and K.-C. Wang, "Review of Internet of Things (IoT) in electric power and energy systems," *IEEE Internet Things J.*, vol. 5, no. 2, pp. 847–870, Apr. 2018.
- [8] T.-S. Hsu *et al.*, "An IoT-based sag monitoring system for overhead transmission lines," in *Proc. IEEE PES GTD Grand Int. Conf. Exposit. Asia (GTD Asia)*, Bangkok, Thailand, Mar. 2019, pp. 515–519.
- [9] P. Musilek, J. Heckenbergerova, and M. M. I. Bhuiyan, "Spatial analysis of thermal aging of overhead transmission conductors," *IEEE Trans. Power Del.*, vol. 27, no. 3, pp. 1196–1204, Jul. 2012.
- [10] E. Lindberg, "The overhead line sag dependence on weather parameters and line current," M.S. thesis, Dept. Earth Sci., Uppsala Univ., Uppsala, Sweden, 2011.
- [11] J. A. Jiang *et al.*, "A novel weather information-based optimization algorithm for thermal sensor placement in smart grid," *IEEE Trans. Smart Grid*, vol. 9, no. 2, pp. 911–922, Mar. 2018.
- [12] *IEEE Standard for Calculating the Current-Temperature of Bare Overhead Conductors*, IEEE Standard 738-2012, 2012.
- [13] P. Zhang, M. Shao, A. R. Leon, D. H. Ramsay, and M. Graham, "Determination of static thermal conductor rating using statistical analysis method," in *Proc. 3rd Int. Conf. Elect. Utility Deregulation Restructuring Power Technol.*, Nanjing, China, Apr. 2008, pp. 1237–1243.
- [14] J. Heckenbergerova, P. Musilek, and K. Filimonenkov, "Assessment of seasonal static thermal ratings of overhead transmission conductors," in *Proc. Power Energy Soc. Gen. Meeting*, Detroit, MI, USA, Jul. 2011, pp. 1–8.
- [15] J. Teh, C. M. Lai, and Y. H. Cheng, "Impact of the real-time thermal loading on the bulk electric system reliability," *IEEE Trans. Rel.*, vol. 66, no. 4, pp. 1110–1119, Dec. 2017.
- [16] D. M. Kim, J. M. Cho, H. S. Lee, H. S. Jung, and J. O. Kim, "Prediction of dynamic line rating based on assessment risk by time series weather model," presented at the Int. Conf. Probab. Methods Appl. Power Syst., Stockholm, Sweden, Jun. 2006.
- [17] J. F. Hall and A. K. Deb, "Prediction of overhead transmission line ampacity by stochastic and deterministic models," *IEEE Trans. Power Del.*, vol. 3, no. 2, pp. 789–800, Apr. 1988.
- [18] M. Matus *et al.*, "Identification of critical spans for monitoring systems in dynamic thermal rating," *IEEE Trans. Power Del.*, vol. 27, no. 2, pp. 1002–1009, Apr. 2012.
- [19] J. Black, S. Connor, and J. Colandairaj, "Planning network reinforcements with dynamic line ratings for overhead transmission lines," in *Proc. 45th Int. Univ. Power Eng. Conf. (UPEC)*, Cardiff, U.K., Aug. 2010, pp. 1–6.
- [20] J. Hosek, P. Musilek, E. Lozowski, and P. Pytlak, "Effect of time resolution of meteorological inputs on dynamic thermal rating calculations," *IET Gener. Transm. Distrib.*, vol. 5, no. 9, pp. 941–947, Sep. 2011.
- [21] P. Pytlak and P. Musilek, "An intelligent weather-based system to support optimal routing of power transmission lines," in *Proc. IEEE Elect. Power Energy Conf. (EPEC)*, Halifax, NS, Canada, Aug. 2010, pp. 1–6.
- [22] *Overhead Electrical Conductors—Calculation Methods for Stranded Bare Conductors*, IEC/TR Standard 61597-1995, 1995.
- [23] *Thermal Behaviour of Overhead Conductors*, CIGRÉ, Paris, France, 2002.
- [24] N. P. Schmidt, "Comparison between IEEE and CIGRE ampacity standards," *IEEE Trans. Power Del.*, vol. 14, no. 4, pp. 1555–1559, Oct. 1999.
- [25] S. Abbott, S. Abdelkader, L. Bryans, and D. Flynn, "Experimental validation and comparison of IEEE and CIGRE dynamic line models," in *Proc. 45th Int. Univ. Power Eng. Conf. (UPEC)*, Cardiff, U.K., Aug. 2010, pp. 1–5.
- [26] F. Xu, "Dynamic thermal rating monitoring and analysis for overhead lines," M.S. thesis, Dept. Elect. Electron. Eng., Manchester Univ., Manchester, U.K., 2014.
- [27] W. C. Ke, B. H. Liu, and M. J. Tsai, "Constructing a wireless sensor network to fully cover critical grids by deploying minimum sensors on grid points is NP-complete," *IEEE Trans. Comput.*, vol. 56, no. 5, pp. 710–715, May 2007.
- [28] M. Naem, U. Pareek, and D. C. Lee, "Swarm intelligence for sensor selection problems," *IEEE Sens. J.*, vol. 12, no. 8, pp. 2577–2585, Apr. 2012.
- [29] K. Worden and A. P. Burrows, "Optimal sensor placement for fault detection," *Eng. Struct.*, vol. 23, pp. 885–901, Aug. 2001.
- [30] T.-H. Yi, H.-N. Li, and M. Gu, "Optimal sensor placement for structural health monitoring based on multiple optimization strategies," *Struct. Design Tall Special Build.*, vol. 20, pp. 881–900, Nov. 2011.
- [31] P. Krishnamurthy and F. Khorrami, "Optimal sensor placement for monitoring of spatial networks," *IEEE Trans. Autom. Sci. Eng.*, vol. 15, no. 1, pp. 33–44, Jan. 2018.
- [32] R. Eberhart and J. Kennedy, "A new optimizer using particle swarm theory," in *Proc. 6th Int. Symp. Micro Mach. Human Sci.*, Nagoya, Japan, Oct. 1995, pp. 39–43.
- [33] A. P. Engelbrecht, *Fundamentals of Computational Swarm Intelligence*. Hoboken, NJ, USA: Wiley, 2006.
- [34] J. Kennedy and R. C. Eberhart, "A discrete binary version of the particle swarm algorithm," in *Proc. IEEE Int. Conf. Syst. Man Cybern. Comput. Cybern. Simulat.*, Orlando, FL, USA, Oct. 1997, pp. 4104–4108.
- [35] Y. Morsly, N. Aouf, M. S. Djouadi, and M. Richardson, "Particle swarm optimization inspired probability algorithm for optimal camera network placement," *IEEE Sens. J.*, vol. 12, no. 5, pp. 1402–1412, May 2012.
- [36] C. Mishra, K. D. Jones, A. Pal, and V. A. Centeno, "Binary particle swarm optimisation-based optimal substation coverage algorithm for phasor measurement unit installations in practical systems," *IET Gener. Transm. Distrib.*, vol. 10, no. 2, pp. 555–562, Feb. 2016.
- [37] J. Yang, H. Zhang, Y. Ling, C. Pan, and W. Sun, "Task allocation for wireless sensor network using modified binary particle swarm optimization," *IEEE Sens. J.*, vol. 14, no. 3, pp. 882–892, Mar. 2014.
- [38] S. Mirjalili and A. Lewis, "S-shaped versus V-shaped transfer functions for binary particle swarm optimization," *Swarm Evol. Comput.*, vol. 9, pp. 1–14, Mar. 2013.
- [39] L. Wang, X. Fu, J. Fang, H. Wang, and M. Fei, "Optimal node placement in industrial wireless sensor networks using adaptive mutation probability binary particle swarm optimization algorithm," in *Proc. 7th Int. Conf. Nat. Comput. (ICNC)*, vol. 4, Shanghai, China, Jul. 2011, pp. 2199–2203.



Joe-Air Jiang (Senior Member, IEEE) received the M.S. and Ph.D. degrees in electrical engineering from National Taiwan University (NTU), Taipei, Taiwan, in 1990 and 1999, respectively.

He is currently a Distinguished Professor with the Department of Biomechatronics Engineering, NTU. He was the Chairman of Taiwan Institute of Biological Mechatronics. He is an Active Researcher. He is currently the Principal Investigator of many large-scale integration projects funded by the Ministry of Science and Technology and the Council of Agriculture of the Executive Yuan, Taipei. He has published 422 papers in different journals and conference proceedings, was granted 65 intellectual patents from the USA and Taiwan, and edited a one book with Springer-Verlag. He and his research team got interviewed by the BBC and Discovery channel, and his research achievements have been broadcasted around the world via BBC and Discovery Channel in 2013 and 2014, respectively. His specialties in power transmission system are computer relaying, solar generation system, fault detection, fault classification, fault location, power quality event analysis, and smart grid system. Besides, his areas of interest are diverse, which cover wireless sensor network (WSN)/Internet-of-Things (IoT) technologies, automatic system for agroecological monitoring with WSN/IoT, biomechatronics, smart grid, solar generation systems, power system, computer relaying, bioeffects of electromagnetic wave, and low-level laser therapy.

Dr. Jiang received the research awards and the best paper awards at various occasions. In 2014, he was awarded by the 38th Ten Outstanding Agriculturist from Taiwan District of Kiwanis International. His achievements on innovation and international collaboration also earned him awards. He received the Outstanding Research Award from the Ministry of Science and Technology in 2020, the Award of Outstanding Electrical Engineering Professor from Chinese Institute of Engineers in 2016, the Lifetime Achievement Award for International Outstanding Inventors from the seventh International Innovation and Invention Competition in 2016, the Award of continuing dedicated service and research contribution from Intel-NTU Connected Context Computing Center in 2015, and the Award of Intel Labs Distinguished Collaborator from Intel in 2016, respectively. He also received the Excellence in Teaching Awards from NTU in 2004, 2012, 2013, 2016, and 2019, the Excellence in Teaching Awards from College of Bioresources and Agriculture at NTU in 2014 and 2015, and the Excellent Mentor Award from NTU in 2011.



Jen-Cheng Wang received the B.S. degree in electronic engineering from Chang Gung University, Tao-Yuan, Taiwan, in 2005, and the Ph.D. degree in bioindustrial mechatronics engineering from National Taiwan University (NTU), Taipei, Taiwan, in 2013.

Since 2016, he has been a Postdoctoral Researcher with the Department of Biomechatronics Engineering, NTU. His current research interests include the data mining and analysis, machine learning and deep learning, smart grid, wireless sensor networks, and optoelectrical characterization of photovoltaic module.



Hung-Shuo Wu received the B.S. and M.S. degrees in bioindustrial mechatronics engineering from National Taiwan University, Taipei, Taiwan, in 2016 and 2018, respectively, where he is currently pursuing the M.S. degree.

His research interests are in the areas of data mining, mathematical optimization, and smart grid.



Chien-Hsing Lee (Senior Member, IEEE) was born in Pingtung, Taiwan, in 1967. He received the B.S. degree in electrical engineering from Arizona State University, Tempe, AZ, USA, in 1993, and the M.S.E.E. and Ph.D. degrees from Georgia Institute of Technology, Atlanta, GA, USA, in 1995 and 1998, respectively.

He is currently a Professor with National Cheng Kung University, Tainan, Taiwan. His research interests include power system analysis and marine mechatronic systems.



Cheng-Ying Chou (Member, IEEE) received the Ph.D. degree in chemical engineering from Rice University, Houston, TX, USA, in 2005.

She worked as a Senior Research Associate with the Illinois Institute of Technology, Chicago, IL, USA, from 2005 to 2007. In 2007, she joined the Department of Biomechatronics Engineering, National Taiwan University, Taipei, Taiwan. Her research interests are focused on the development and investigation of bioimaging techniques and developing optimization algorithms and applying machine learning methods for power dispatch.



Li-Cheng Wu (Senior Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from the National Taiwan University of Science and Technology, Taipei, Taiwan, in 1997 and 1999, respectively, and the Ph.D. degree from National Taiwan University, Taipei, in 2013.

He is currently a Deputy General Manager with Taiwan Power Research Institute, New Taipei City, Taipei. His main research interests are real-time simulation, IEC61850 applications, power electronics, high-voltage test, and power system protection.



Yu-Cheng Yang was born in New Taipei City, Taiwan, in 1988. He received the M.S. degree from the Institute of Electro-Optical Engineering, National Taiwan University, Taipei, Taiwan, in 2012, where he is currently pursuing the Ph.D. degree in biomechatronics engineering.

His research interests are in the areas of automatic control, computer science, wireless sensor network, and smart grid.

Dr. Yang was a recipient of the Chinese Institute of Engineer EE Group's Excellent Award in 2010.